DIGITAL TWIN CHALLENGE 2026

GlucoTwin

A patient model.
An accountable next step.

alpha · vnrvjiet
Shanmukh sai gara · registered team size 1

Public / simulated data · Synthetic narration · Doctor review pending

A patient model connected to a clinic review workflow | Adult Type 2 diabetes · staffed monitoring | Research prototype · doctor review pending

Clinic review workflow

01 **Identify a patient needing review**

02 Inspect context, prediction and reliability

03 Document follow-up and compare the outcome

One review episode connects prediction to documented follow-up.

Challenge alignment

01 **Historical context plus incoming CGM**

02 An evolving state and future-excursion model

03 A doctor-facing review dashboard

Published requirements guide the build; numerical judging weights are not published.

Measured CGM at native resolution

01 **100 patients · 109 recordings**

02 112,462 usable measured observations

03 15-minute sampling; gaps preserved

Native 15-minute observations preserve the timing evidence.

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04 | Measured CGM at native resolution

100 patients · 109 recordings | 112,462 usable measured observations | 15-minute sampling; gaps preserved

Primary source

Historical context and availability

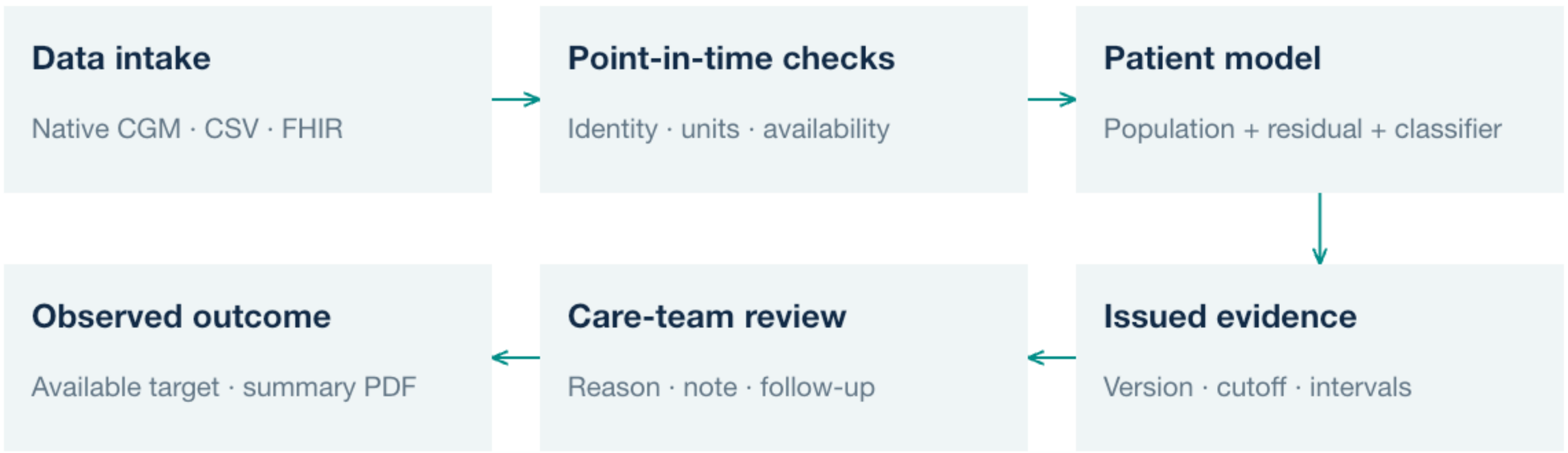
01 **Use information available at the prediction time**

02 Record source, observed time and available time

03 Exclude untimed labs and medications

Late information cannot rewrite an earlier forecast.

System architecture



SQLite persists observations, context, forecasts, outcomes and review actions.

Human actions and model evidence are stored separately.

Population forecasting foundation

01 **Gradient boosting over causal CGM features**

02 Historical context enters the candidate model

03 Persistence, trend and context-free comparisons

Context is an input; its benefit must be measured.

Online patient adaptation

01 **Bounded, shrunk residual correction**

02 Update only after an earlier target matures

03 Compare with the identical unadapted model

Residual adaptation is bounded and learns only from available outcomes.

Bounded, shrunk residual correction | Update only after an earlier target matures | Compare with the identical unadapted model

Future-excursion prediction

01 **Any reading above 180 mg/dL within the horizon**

02 30 minutes primary; 60 minutes separate

03 Ongoing high glucose is displayed separately

Forecast a future event; distinguish ongoing high glucose.

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Any reading above 180 mg/dL within the horizon | 30 minutes primary; 60 minutes separate | Ongoing high glucose is displayed separately

Calibration and alert burden

01 **Separate probability and interval calibration**

02 Validation-selected operating thresholds

03 Report false alarms, misses and abstention

An engineering alert budget awaits clinician review.

Patient-separated internal validation

01 **Five outer patient folds**

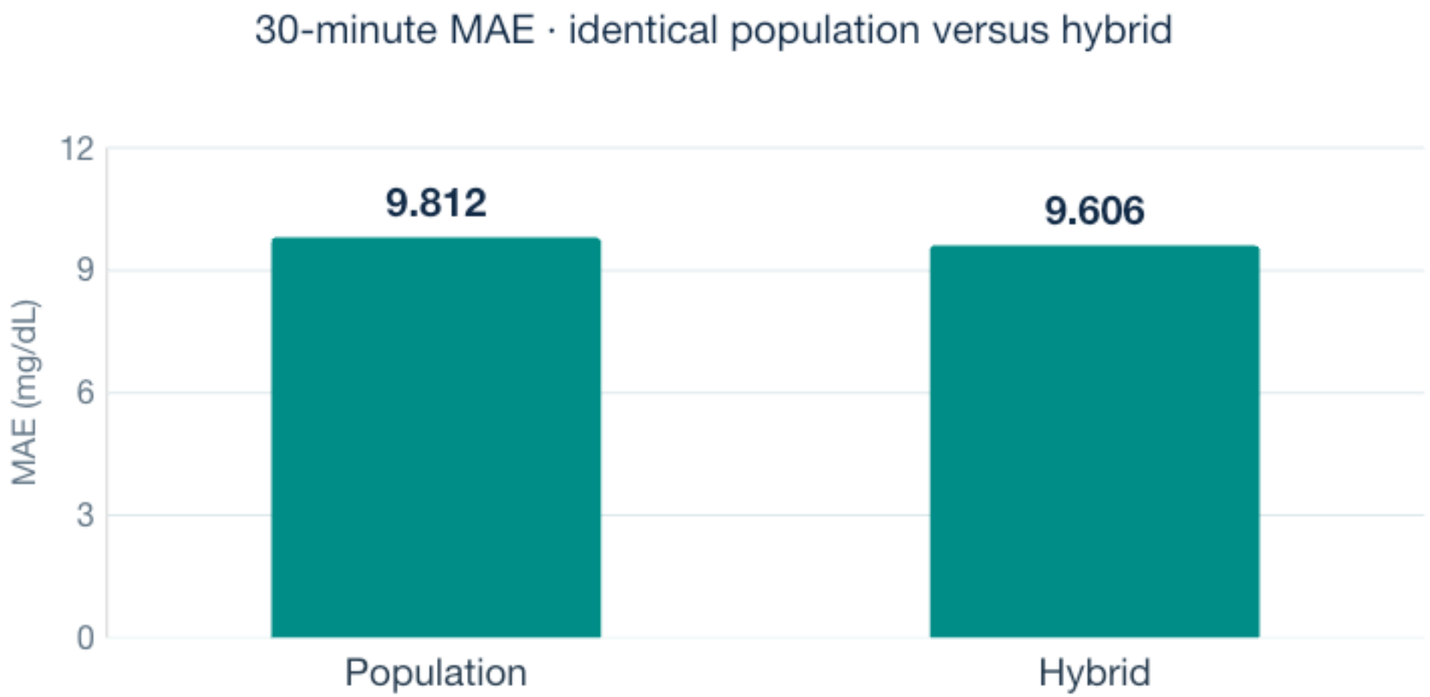
02 Grouped inner selection and separate calibration

03 Out-of-fold results and patient uncertainty

Every patient is evaluated by a model fitted on other patients.

Five outer patient folds | Grouped inner selection and separate calibration | Out-of-fold results and patient uncertainty

Measured model evidence



PAIRED DIFFERENCE

–0.206 mg/dL

95% patient-bootstrap CI
[–0.336, –0.110]

Precision 62.96%
Overall recall 76.06%
False alerts 0.919 / 24 h*

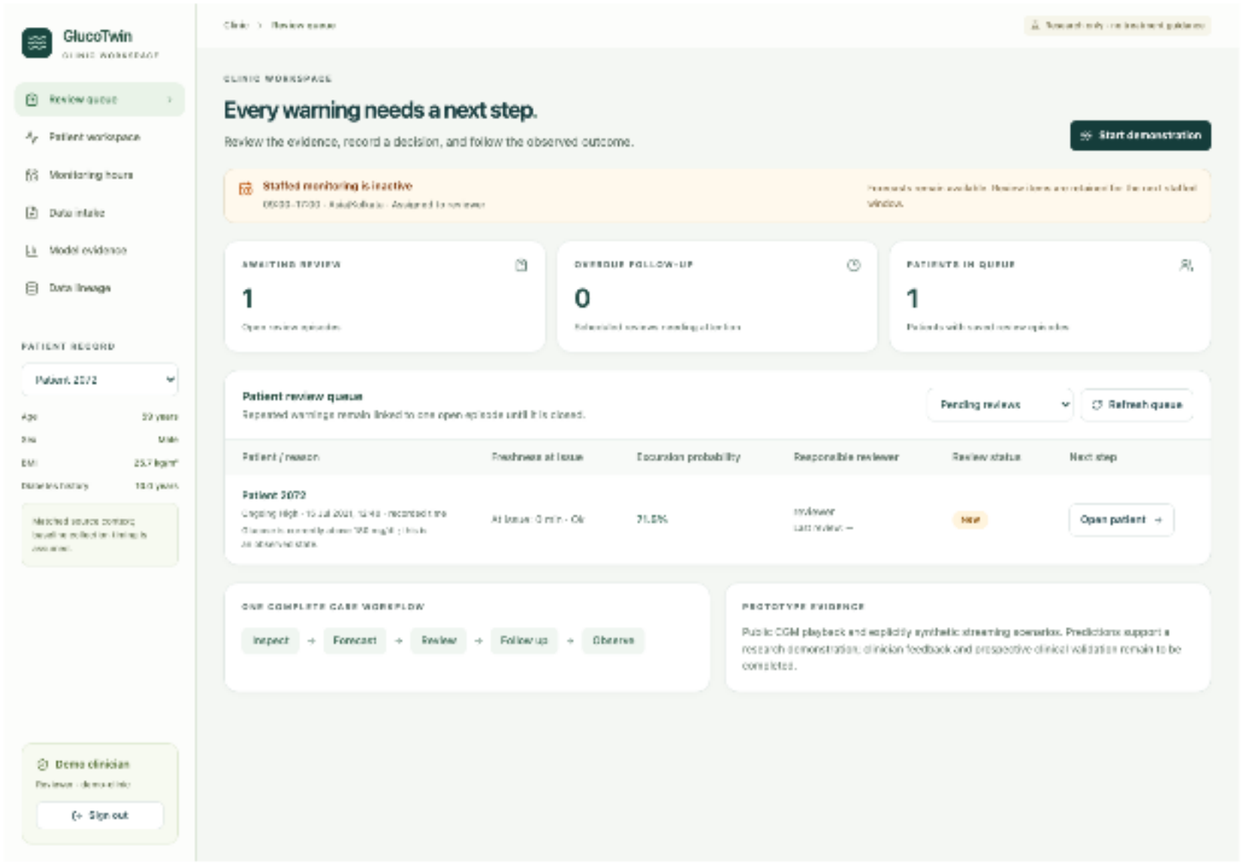
*All eligible forecast time, not staffed hours. Historical-context benefit inconclusive.

A small internal gain does not establish clinical effectiveness.

30-minute MAE: population 9.812 □ hybrid 9.606 | Paired change −0.206; 95% CI [−0.336, −0.110] | Small internal gain; clinical benefit unestablished

Patient review queue

- Risk, freshness, ownership and review status
- Forecast and missing-data concerns distinguished
- Open the patient evidence directly



Actual local application capture

Freshness at issue is distinct from a live sensor connection.

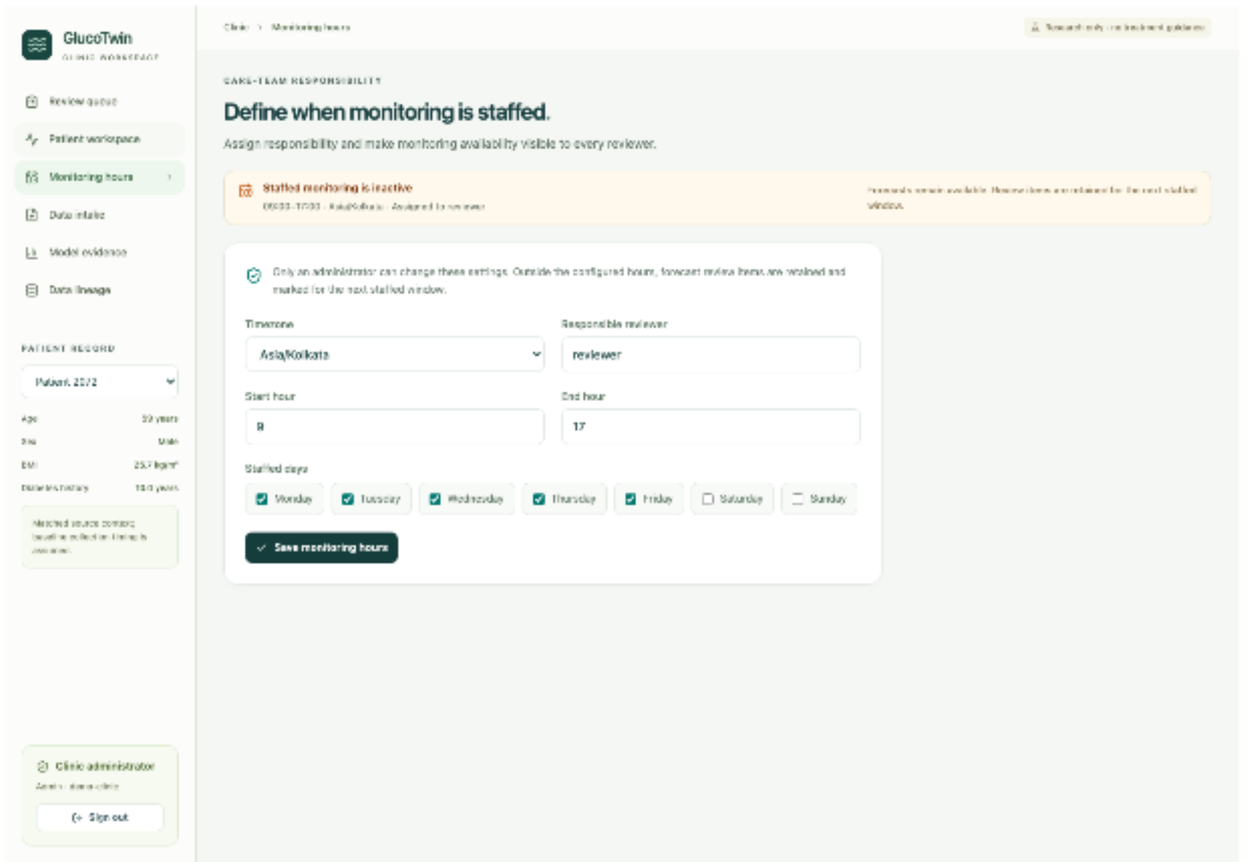
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13 | Patient review queue

Risk, freshness, ownership and review status | Forecast and missing-data concerns distinguished | Open the patient evidence directly

Staffed monitoring windows

- Configured hours and assigned reviewer
- Visible active or inactive status
- Overdue review is an operational concern

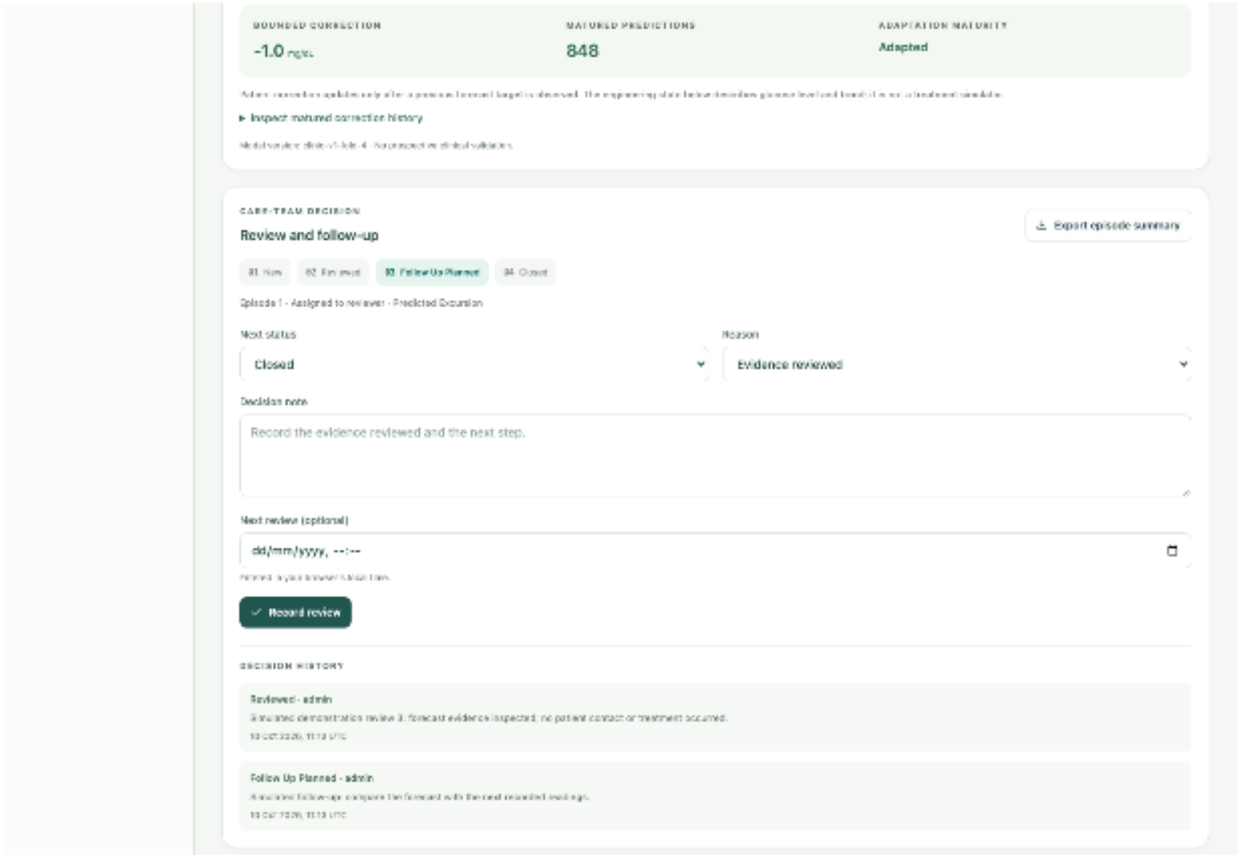


Actual local application capture

Outside hours, items are retained with monitoring visibly inactive.

Persistent review and follow-up

- New → reviewed → follow-up planned → closed
- Reviewer, time, reason, note and next review
- Action history survives restart



Actual local application capture

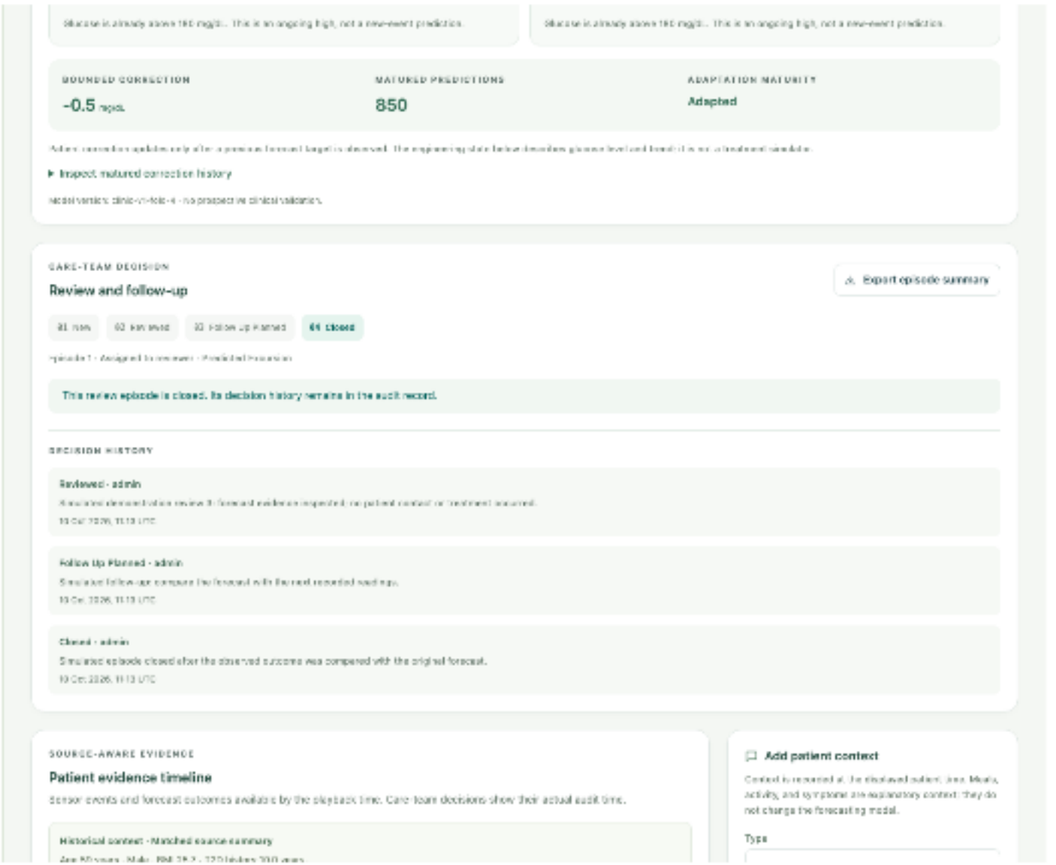
Acknowledgement changes workflow status, never forecast accuracy.

15 | Persistent review and follow-up

New → reviewed → follow-up planned → closed | Reviewer, time, reason, note and next review | Action history survives restart

Patient evidence timeline

- Glucose, context, predictions and outcomes
- Measured, synthetic and self-reported labels
- Late data retains its true availability time



Actual local application capture

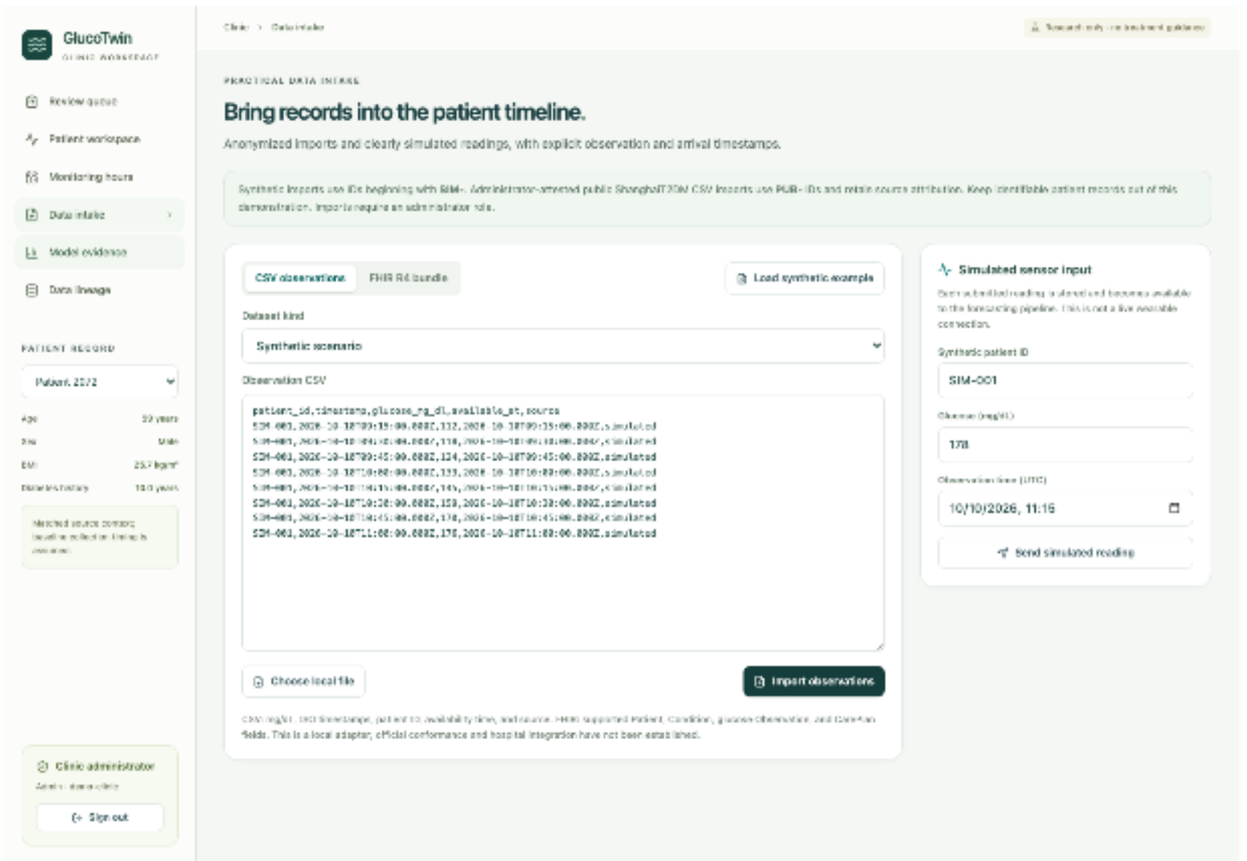
The issue time and input cutoff reconstruct what was known.

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Glucose, context, predictions and outcomes | Measured, synthetic and self-reported labels | Late data retains its true availability time

Practical data intake

- Anonymized CSV and synthetic stream
- Limited Patient / Condition / Observation / CarePlan FHIR
- Identity, units, times and duplicate checks



Actual local application capture

Local FHIR intake is a prototype; conformance is unestablished.

17 | Practical data intake

Anonymized CSV and synthetic stream | Limited Patient / Condition / Observation / CarePlan FHIR | Identity, units, times and duplicate checks

Primary source

Consultation summary

01 **Observed patterns and data coverage**

02 Forecasts versus outcomes and unresolved reviews

03 Clinician notes and provenance

Export the same evidence the reviewer inspected.

Access, audit and recovery

01 **Reviewer/admin roles and clinic access**

02 Environment secrets and persistent audit

03 Consistent backup and offline restore

Local controls work; clinical deployment controls remain separate gates.

Clinical pilot gates

01 **Doctor workflow review remains pending**

02 Verify authorized source latency and Indian applicability

03 Shadow evaluation before alerts influence care

Shadow evaluation precedes any influence on care.

Doctor workflow review remains pending | Verify authorized source latency and Indian applicability | Shadow evaluation before alerts influence care

Reproducibility and handoff

01 **Source, setup, frozen protocol and benchmark**

02 Architecture, presentation and demo assets

03 Team alpha · vnrvjiet · Shanmukh sai gara

Registration size 1; four coordinated delivery workstreams.